Lecture 12 (6)

TSBB[03]6 Multidimensional Signal (|and Image)Analysis

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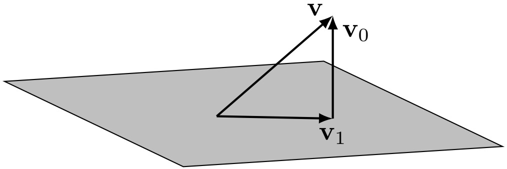
Subspace Representation

We know from previous lectures how to compute the least-squares approximation $\mathbf{v}_1 \in U \subset V$ of a vector $\mathbf{v} \in V$:



Given a subspace basis $\mathbf{B} \in \mathbb{R}^{n\times m}$ and scalar product $\mathbf{G}_0$, the approximation that minimises the norm $\|\mathbf{v}-\mathbf{v}_1\|$ is given by $\mathbf{v}_1 = \mathbf{B}(\mathbf{B}^*\mathbf{G}_0\mathbf{B})^{-1}\mathbf{B}^*\mathbf{G}_0\mathbf{v}$, which is independent of $\mathbf{B}$ (but not of U or $\mathbf{G}_0$).

What if the Subspace is not Given?

Previously, the subspace was assumed to be given at the outset.

Another viewpoint, which we will consider today, is to take $\mathbf{v}$ to be stochastic (with known statistics), and try to find the subspace of dimension m which *on average* captures most of the signal.

More precisely, we seek an m-dimensional subspace U that minimises the expected value of the difference, i.e.

$$\varepsilon = \mathbb{E}\left[\|\mathbf{v} - \mathbf{v}_1\|^2\right].$$

This is computed over the whole distribution of $\mathbf{v}$!

Initial Simplifications

To simplify matters, we make the following assumptions:

- The vector space is real valued and of finite dimension, i.e. $V = \mathbb{R}^n$.
- The scalar product is $\langle \mathbf{u} \,|\, \mathbf{v} \rangle = \mathbf{v}^\top \mathbf{u}$, i.e. we have $\mathbf{G}_0 = \mathbf{I}$.
- We only care about finding an ON-basis $\mathbf{B}$ of U, i.e. $\mathbf{B}^\top \mathbf{B} = \mathbf{I}$ and $\tilde{\mathbf{B}} = \mathbf{B}$.
    - However, if $m < n$, we do not have $\mathbf{B}\mathbf{B}^\top = \mathbf{I}$.

With these assumptions in force, the least-squares approximation of $\mathbf{v}$ in the subspace will be $\mathbf{v}_1 = \mathbf{B}\mathbf{c} = \mathbf{B}\mathbf{B}^\top \mathbf{v}$, and we therefore seek to minimize

$$\varepsilon = \mathbb{E}\left[\|\mathbf{v} - \mathbf{B}\mathbf{B}^\top \mathbf{v}\|^2\right]$$

subject to $\mathbf{B}^\top \mathbf{B} = \mathbf{I}$. Equivalently: maximise $\varepsilon_1 = \mathbb{E}\left[\mathbf{v}^\top \mathbf{B}\mathbf{B}^\top \mathbf{v}\right]$.

Example: 1D Subspace

Assume we are looking for a one-dimensional subspace, i.e. $\mathbf{B}$ consists of only one column $\mathbf{b}_1$.

Then

$$\varepsilon_1 = \mathbb{E}\big[\mathbf{v}^\top \mathbf{B}\mathbf{B}^\top \mathbf{v}\big] = \mathbb{E}\big[\mathbf{v}^\top \mathbf{b}_1 \mathbf{b}_1^\top \mathbf{v}\big] = \mathbb{E}\big[\mathbf{b}_1^\top \mathbf{v}\mathbf{v}^\top \mathbf{b}_1\big] = \mathbf{b}_1^\top \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_1.$$

With the constraint $g = \|\mathbf{b}_1\|^2 = 1$, *Lagrange's method of multipliers* tells us that we must have

$$\nabla\varepsilon_1 = \lambda\nabla g \iff \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_1 = \lambda\mathbf{b}_1$$

at the optimum. Then

$$\varepsilon_1 = \mathbf{b}_1^\top \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_1 = \lambda\mathbf{b}_1^\top\mathbf{b}_1 = \lambda.$$

Conclusion: We should choose $\mathbf{b}_1$ as the eigenvector of $\mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]$ with the largest eigenvalue!

The Correlation Matrix

In the 1D example, we made use of the *correlation matrix* $\mathbf{C} = \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]$. This matrix will be important in the general case too!
Note:

- The correlation matrix is real and symmetric, meaning all eigenvalues are real.
- The correlation matrix is positive semi-definite, i.e. $\mathbf{u}^\top \mathbf{C}\mathbf{u} \geq 0$ for all $\mathbf{u}$.
- The correlation matrix can be estimated from samples of the signal:

$$\mathbf{C} \approx \frac{1}{p} \sum_{k=1}^{p} \mathbf{v}_k \mathbf{v}_k^\top .$$

Example: 2D Subspace

We are now looking for a two-dimensional subspace, i.e. $\mathbf{B} = \begin{pmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{pmatrix}$.

Issue: There are infinitely many ON subspace bases — $\mathbf{B}$ is not unique.

Solution: Require that $\mathbf{B}^\top \mathbf{C} \mathbf{B}$ is diagonal.

- This means that the subspace basis vectors are *uncorrelated*, in addition to just being orthogonal and normalised!

Example: 2D Subspace (contd.)

Now

$$\varepsilon_1 = \mathbb{E}\big[\mathbf{v}^\top \mathbf{B}\mathbf{B}^\top \mathbf{v}\big] = \mathbb{E}\big[\mathbf{b}_1^\top \mathbf{v}\mathbf{v}^\top \mathbf{b}_1 + \mathbf{b}_2^\top \mathbf{v}\mathbf{v}^\top \mathbf{b}_2\big] = \mathbf{b}_1^\top \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_1 + \mathbf{b}_2^\top \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_2.$$

Using the constraints $g_1 = \|\mathbf{b}_1\|^2 = 1$ and $g_2 = \|\mathbf{b}_2\|^2 = 1$ with Lagrange's method of multipliers, the gradients need to be linearly dependent, i.e.

$$\nabla\varepsilon_1 = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 \iff \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_1 + \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_2 = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2$$

at the optimum. Then, since $\langle \mathbf{b}_1 \,|\, \mathbf{b}_2 \rangle = 0$ and since the basis vectors are uncorrelated,

$$\varepsilon_1 = \mathbf{b}_1^\top \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_1 + \mathbf{b}_2^\top \mathbb{E}\big[\mathbf{v}\mathbf{v}^\top\big]\mathbf{b}_2 = \lambda_1 \mathbf{b}_1^\top \mathbf{b}_1 + \lambda_2 \mathbf{b}_2^\top \mathbf{b}_2 = \lambda_1 + \lambda_2.$$

Conclusion: We should choose $\mathbf{b}_1$ and $\mathbf{b}_2$ as eigenvectors of $\mathbf{C}$ with the largest eigenvalues!

Conclusions

Using the same idea, it is easy to generalise to any $m \leq n$.
Note: If $m = n$, this means that $\varepsilon_1 = \lambda_1 + \ldots + \lambda_n$, i.e. the sum of all eigenvalues of $\mathbf{C}$!

This means that, for any $m \leq n$, we have $\varepsilon = \lambda_{m+1} + \ldots + \lambda_n$. The interpretation is that with an m-dimensional subspace, the expected approximation error is ε.

The eigenvectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ of $\mathbf{C}$ are called the *principal components* of the signal, and $\lambda_1, \ldots, \lambda_n$ are (unfortunately) called their corresponding *magnitudes*.

How to Choose m?

When choosing the size of the subspace, there are many valid strategies and considerations:

- If we *know* the signal subspace dimension, we should choose m to be that. This suppresses noise, which is found in all the other dimensions.
- If the manager tells us that our budget is m, and we don't know anything about the signal, we choose m. This gives the smallest expected approximation error.
- If the requirement is to capture, e.g., at least 95% of the signal *on average*, we choose m such that $\frac{\lambda_1+\ldots+\lambda_m}{\lambda_1+\ldots+\lambda_n} \geq 0.95$.
- We can consider the quotients $\frac{\lambda_{k+1}}{\lambda_k}$. When this becomes small, the benefit of including the next dimension is small.

Further Consideration

A number of additional considerations can be of relevance in principal component analysis:

- It is often useful to centre the data! Subtract the centroid, then form $\mathbf{C}$, then use PCA to analyse the signal around the centroid.
- For numerical purposes, it is better to use the SVD instead of eigenvalue decomposition. If the signal observations are given as

$$\mathbf{W} = \begin{pmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_p \end{pmatrix},$$

  The eigenvectors of $\mathbf{C} = \mathbf{W}\mathbf{W}^\top$ will be the left singular vectors of $\mathbf{W}$, and the eigenvalues will be the squares of the singular values! This avoids squaring $\mathbf{W}$.
- PCA is often applied to *feature embeddings* $\mathbb{R}^n \ni \mathbf{v}_j \mapsto \phi(\mathbf{v}_j) \in \mathbb{R}^d$ (the *kernel trick*).

Some Applications of PCA

PCA has many applications, among others:

- Noise reduction: If we know the signal lives in a certain subspace, we can remove all noise in directions orthogonal to the subspace.
- Dimensionality reduction: By finding an appropriate subspace to analyse the signal in, it may be easier to do the analysis.
- Compression (see lab): By only using the m most significant principal components to represent the data, we can reduce storage or transmission costs.
  - Since the compression is data dependent, it can be efficient.
  - Since $\mathbf{B}$ needs to be stored/transmitted, there will be some overhead.